Polynomials – 5

Integer coefficients, divisibility and irreducibility

8 сентября 2026 года

Theoretical minimum.

Theorem 1 (Gauss). A polynomial in $\mathbb{Z}[x]$ which is a product of two non-constant polynomials in $\mathbb{Q}[x]$ is a product of two non-constant polynomials in $\mathbb{Z}[x]$ of the same degrees. The content is multiplicative, so a primitive polynomial times a primitive polynomial is primitive.

Theorem 2 (the three criteria). Eisenstein: if a prime p divides all coefficients but the leading one and $p^2 \nmid a_0$, then f is irreducible over $\mathbb{Q}$. Reduction: if f stays of the same degree and becomes irreducible modulo some prime, then f is irreducible. Perron: if $|a_{n-1}| > 1 + |a_{n-2}| + \cdots + |a_0|$ and $a_0 \neq 0$, then f is irreducible.

Fact 1 (values control the polynomial). For $f \in \mathbb{Z}[x]$ and integers $a \neq b$ one has $a - b \mid f(a) - f(b)$; a polynomial taking the value ± 1 at too many integers is forced to be small; and if f takes a prime value at many integers, its factorisations are severely restricted.

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1. Prove Gauss's lemma, Eisenstein's criterion and the reduction criterion, and use them to prove that $1 + x + \cdots + x^{p-1}$ is irreducible over $\mathbb{Q}$ for prime p.

2. (Высшая лига 2020) Can the polynomial $(x-1)(x-2)\cdots(x-n)-1$ be written as a product of two polynomials of smaller degree with integer coefficients for some n?

3. (IMC 2012) Let a be a rational number and n a positive integer. Prove that the polynomial $X^{2^n}(X+a)^{2^n} + 1$ is irreducible in $\mathbb{Q}[X]$.

4. (IMC) Let f be a polynomial of degree 2 with integer coefficients such that $f(k)$ is divisible by 5 for every integer k. Prove that all coefficients of f are divisible by 5, and find the analogous statement for degree n and a prime p.

5. (IMC) Let f be a real polynomial of degree n such that $\frac{f(k)-f(m)}{k-m}$ is an integer for all integers $0 \leqslant k < m \leqslant n$. Prove that $a - b$ divides $f(a) - f(b)$ for every pair of distinct integers a and b.